

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Systemic Lupus Erythematosus (SLE) (POETYK SLE-1) **Short Title/Acronym:** POETYK SLE-1 **Protocol Number:** IM011-246 **NCT Number:** NCT05617677 **Trial Status:** Recruiting **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** Deucravacitinib (Trade Name: Sotyktu) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effectiveness (efficacy) and safety of deucravacitinib compared with placebo in a patient population with active, moderate to severe Systemic Lupus Erythematosus (SLE). **Secondary Objectives:** To evaluate the proportion of participants achieving British Isles Lupus Assessment Group-based Combined Lupus Assessment (BICLA) response, dual SRI(4) and BICLA responders, Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) response, Lupus Low Disease Activity State (LLDAS), reduction in active joint count, and improvement in patient-reported fatigue.

2.2 Background & Rationale Systemic lupus erythematosus (SLE) is a chronic, multi-organ autoimmune disease associated with a high burden of illness and a need for targeted therapies that do not rely heavily on glucocorticoids. Deucravacitinib is a first-in-class, oral, selective, allosteric tyrosine kinase 2 (TYK2) inhibitor. TYK2 mediates signaling of Type I interferons, IL-12, and IL-23, which are key cytokines involved in SLE pathophysiology. Following successful Phase 2 results (the PAISLEY trial) demonstrating significant clinical and cutaneous symptom improvements, POETYK SLE-1 aims to confirm the efficacy and safety of deucravacitinib, alongside a protocolized glucocorticoid-tapering strategy, in a larger global population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant, Investigator) **Randomization:** 3:2 (Deucravacitinib to Placebo)

3.2 Planned Enrollment & Duration Target **NS:** 490 participants **Number of Sites:** Global multi-center trial (North and South)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults aged 18 to 75 years old diagnosed with Systemic Lupus Erythematosus (SLE) at least 24 weeks before the screening visit. 2. Meet the European Alliance of Associations for Rheumatology (EULAR)/American College of Rheumatology (ACR) 2019 classification criteria for SLE. 3. One of the following: positive antinuclear antibodies (ANA) $\geq$ 1:80 at screening, OR positive anti-dsDNA, OR positive anti-Smith (anti-Sm) as determined by the central laboratory at screening. 4. Total Systemic Lupus Erythematosus Disease Activity Index-2K (SLEDAI-2K) score $\geq$ 6 points and clinical SLEDAI-2K score $\geq$ 4 points with joint involvement, and/or cutaneous vasculitis, and/or rash. (Note: Lupus headache, alopecia, organic brain syndrome, and mucosal ulcers must be recorded but do not count toward the points required for screening at entry). 5. At least one SLE background therapy (immunosuppressant and/or antimalarial) is required for $\geq$ 12 weeks before the screening visit, must be at a stable dose for $\geq$ 8 weeks before the screening visit, and must remain stable until randomization and throughout study participation. 6. Oral corticosteroid (OCS; prednisone or equivalent) background therapy is permitted but not required. For participants taking OCS, the dose must be stable for $\geq$ 2 weeks before the screening visit, cannot exceed 30 mg/day at screening, and must remain stable until the Week 4 visit. Participants can be on an OCS as well as an antimalarial and/or an immunosuppressant.

4.2 Exclusion Criteria 1. Diagnosis of drug-induced SLE rather than idiopathic SLE. 2. Other autoimmune diseases (e.g., multiple sclerosis, psoriasis, inflammatory bowel disease) are excluded. (Note: Participants with type I autoimmune diabetes mellitus, thyroid autoimmune disease, Celiac disease, or secondary Sjögren's syndrome are not excluded). 3. SLE overlap syndromes including, but not limited to, rheumatoid arthritis, scleroderma, and mixed connective tissue disease are excluded. 4. Active or unstable lupus neuropsychiatric manifestations, including, but not limited to, any condition defined by BILAG A criteria. 5. Active, severe Class III, and IV, lupus nephritis that requires or may require treatment with cytotoxic agents or high-dose CS. 6. History of congenital or acquired immunodeficiency. 7. Known active infection, or any major episode of infection requiring hospitalization or treatment with parenteral (intramuscular or IV) antimicrobial agents within 30 days of randomization, or treatment with oral antimicrobial agents within 2 weeks of randomization. 8. Currently on any therapy for chronic infection (e.g., pneumocystis, herpes zoster, cytomegalovirus, invasive bacterial or fungal infections, or atypical mycobacteria). 9. Taking more than 1 immunosuppressant at screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Deucravacitinib (specified dose) administered on specified days. **Route of Administration:** Oral **Duration of Treatment:** 52 weeks of double-blind treatment, with the option to continue in a 104-week open-label extension phase.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background standard-of-care treatments (antimalarials, immunosuppressants). Oral corticosteroids (OCS) up to 30 mg/day (stable) are subject to a protocol-instructed tapering strategy to a threshold dose level during the double-blind period. **Prohibited:** Taking more than 1 immunosuppressant at screening; current therapy for chronic infections (e.g., invasive bacterial/fungal, atypical mycobacteria).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent, Medical History, Background Therapy Verification
Baseline	Day 0	Randomization, First Dose, Disease Activity Assessments
Interim	Q4W to Q12W	Safety Labs, Efficacy Assessments (SRI-4, BICLA, CLASI), Glucocorticoid Taper Tracking
End of Study	Week 52	Primary Endpoint Assessment, Final Vital Signs, IP Accountability, Transition to Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants who achieve Systemic Lupus Erythematosus Responder Index-4 [SRI(4)] response. **Measurement Method:** Clinical adjudication utilizing composite disease activity indices (SLEDAI-2K, BILAG, and Physician Global Assessment).

7.2 Secondary & Exploratory Endpoints

- Proportion of participants who achieve British Isles Lupus Assessment Group-based Combined Lupus Assessment (BICLA) response.
- Proportion of participants who achieve both SRI(4) and BICLA (dual responders).
- Proportion of participants with a Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) activity score ≥ 10 at baseline who achieve a CLASI response, defined as a decrease of $\geq 50\%$ from baseline CLASI activity score.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical assessments and patient reporting (including risks typically associated with immunomodulation and TYK2 inhibitors, such as infections). **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to review ongoing safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for primary and secondary efficacy evaluations; Safety Set for all subjects who receive at least one dose of the investigational product. **Sample Size Justification:** Powered to detect a statistically significant difference in SRI(4) response rates between the deucravacitinib arm and the placebo arm ($\alpha = 0.05$, $\text{Power} = 80\%$) based on effect sizes observed in the Phase 2 PAISLEY trial. **Handling of Missing Data:** Non-responder imputation (NRI) for binary composite endpoints (SRI-4, BICLA) where treatment discontinuation or missing data equates to non-response.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring incorporating onsite visits and centralized data review to ensure protocol adherence and patient safety. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing 21 CFR Part 11-compliant systems with automated data clarification forms.

11. Study Identifiers & Governance

Primary ID: IM011-246 **Registry Number:** NCT05617677 **Posting ID:** 708225723703505783 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** POETYN SLE-1 (A Study to Assess Effectiveness and Safety of Deucravacitinib Compared With Placebo in Participants With Active Systemic Lupus Erythematosus (SLE)) **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants With Active Systemic Lupus Erythematosus (SLE)

12. Clinical Framework (Systemic Lupus Erythematosus Specific)

Primary Condition: Active Systemic Lupus Erythematosus (SLE)
Diagnosis Threshold: Active moderate-to-severe disease on background standard-of-care
Medical Methodology: Selective, allosteric tyrosine kinase 2 (TYK2) inhibition
Optimization Target: Achievement of SRI-4 response and successful glucocorticoid-tapering

13. Study Procedures & Methodology

Management Pathway: Standard of care for SLE supplemented with experimental TYK2 inhibitor.
Education Protocol (HCP): Training on protocolized disease activity adjudication (SLEDAI, BILAG, CLASI).
Education Protocol (Patient): Education on self-administration, adverse event reporting, and recording of Patient-Reported Outcomes (PROs).
Quality Audit Mechanism: Independent disease activity adjudication and central monitoring of corticosteroid tapering logs.

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (double-blind) followed by optional 104-week open-label extension
Onsite Visit Cadence: Regular intervals (e.g., Weeks 4, 16, 24, 32, 40, 52)
Remote Monitoring Frequency: Routine oversight per protocol-specific clinical monitoring plan.
Checklist Review Interval: Regular review of patient-reported fatigue, joint counts, and medication compliance.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				